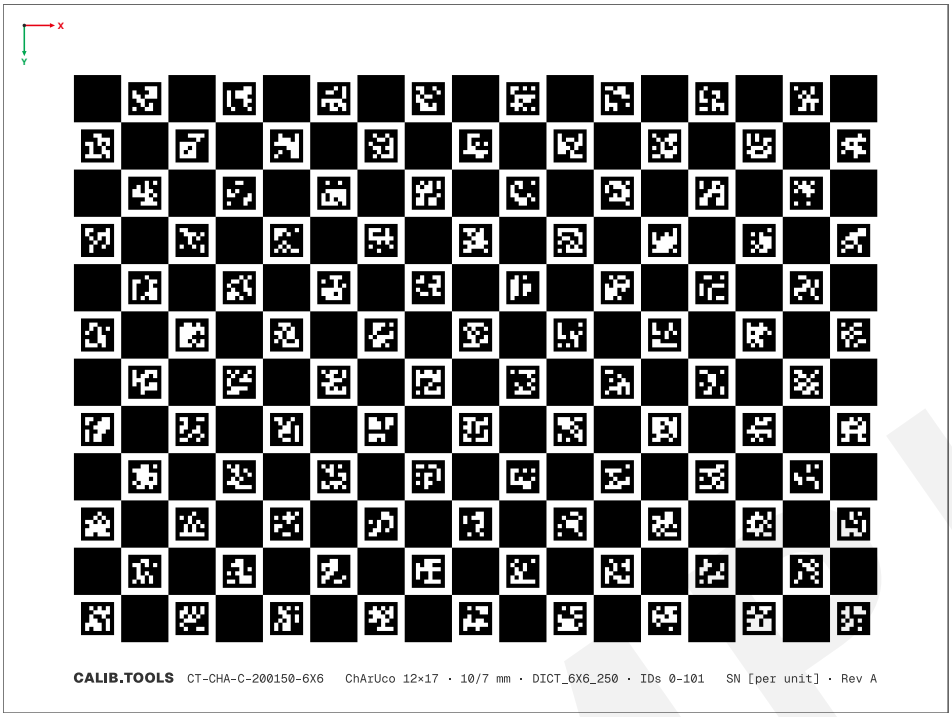
DATASHEET

ChArUco calibration target · ceramic

A rigid technical-ceramic ChArUco target for precise camera calibration where dimensional stability, sharp features and durable handling matter.



Rendering · 200 × 150 mm variant

COORDINATE SYSTEM



(0, 0) · pattern outer corner

X right · Y down
OpenCV pattern frame

1 Ordering

PRINT VERSION · prices online

Part number	Size W × H (mm)	Squares (r × c)	Square / marker (mm)	Dictionary options	Corners	Markers	Weight (kg)
CT-CHA-C-100075-6X6	100 × 75	9 × 14	5 / 3.5	DICT_6X6_250, DICT_6X6_1000	104	63	0.12
CT-CHA-C-150100-6X6	150 × 100	7 × 12	10 / 7	DICT_6X6_250, DICT_6X6_1000	66	42	0.23
CT-CHA-C-200150-6X6	200 × 150	12 × 17	10 / 7	DICT_6X6_250, DICT_6X6_1000	176	102	0.47
CT-CHA-C-300200-6X6	300 × 200	17 × 27	10 / 7	DICT_6X6_250, DICT_6X6_1000	416	229	0.94

All variants: 15 mm white margin · marker side 0.7 × square side · technical ceramic, 4.0 mm · density 3.9 g/cm³.

Prices: calib.tools/targets

2 Key specifications

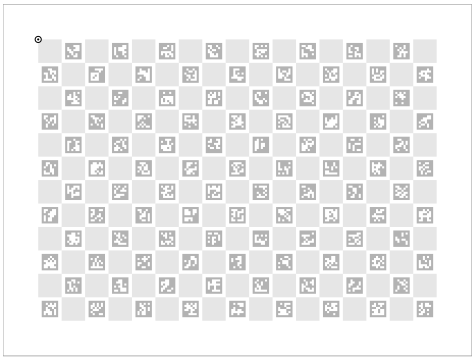
ACCURACY	SUBSTRATE	SIZES	FLATNESS
±2 µm feature position · 20 °C	Technical ceramic 4.0 mm	100 × 75 to 300 × 200 mm	≤ 20 µm 200 × 150 mm

SAMPLE

Unless stated otherwise, dimensions are nominal at 20 °C. Specifications apply to the product family; per-unit results are issued in the inspection report.

1 Specifications · geometry

FEATURE	SPECIFICATION	CONDITION / NOTE
Feature position accuracy	±2 µm	At 20 °C; ChArUco corner positions.
Edge definition	Sharp black / white transition	≤ 20 µm nominal 10–90 % transition width.
Orthogonality	90° ± 0.002°	Pattern axes.
Start ID / placement	0 · row-major	Black square top-left; marker IDs advance through white squares.
OpenCV pattern convention	OpenCV ≥ 4.6 layout	12 rows (even). Older-board layouts require `CharucoBoard.setLegacyPattern(True)`.



Geometry reference · pattern envelope shown.

PATTERN CONVENTION

Reference pattern	12 × 17 squares · 10 / 7 mm.
Marker IDs	0–101, row-major through white squares.
Coordinate frame	Origin at the top-left outer pattern corner; X right, Y down.
Version control	Current ≥ 4.6 convention for an even-row board; no legacy pattern setting.

2 Specifications · physical

FEATURE	SPECIFICATION
Substrate	Technical ceramic (alumina-based).
Pattern process	Fired mineral-pigment screen print; matte black / white pattern.
Thickness	4.0 mm nominal.
Flatness	≤ 20 µm · 200 × 150 mm variant.
Surface	Ra ≤ 0.8 µm nominal; non-glare finish.
Edges	0.5 × 45° chamfer on all edges.
Weight	0.12–0.94 kg, by size; 200 × 150 mm: 0.47 kg.
Mounting	No holes. Optional fixture supports the target at the perimeter.

3 Specifications · environment

FEATURE	SPECIFICATION
Operating temperature	5–40 °C; allow target and camera to stabilise.
Storage temperature	–20–60 °C; store clean, dry and protected.
Thermal expansion	Approx. 7 ppm/K (alumina); use at stated reference temperature for accuracy.
Humidity	10–85 % RH, non-condensing.
Chemical resistance	Resists water and 70 % IPA at ambient conditions; avoid acids, alkalis and abrasives.
UV / outdoor life	Indoor use. No outdoor service-life rating is specified.
Cleaning	Clean with a lint-free cloth; use deionised water or 70 % IPA. Do not scour.

SAMPLE

3 Deliverables

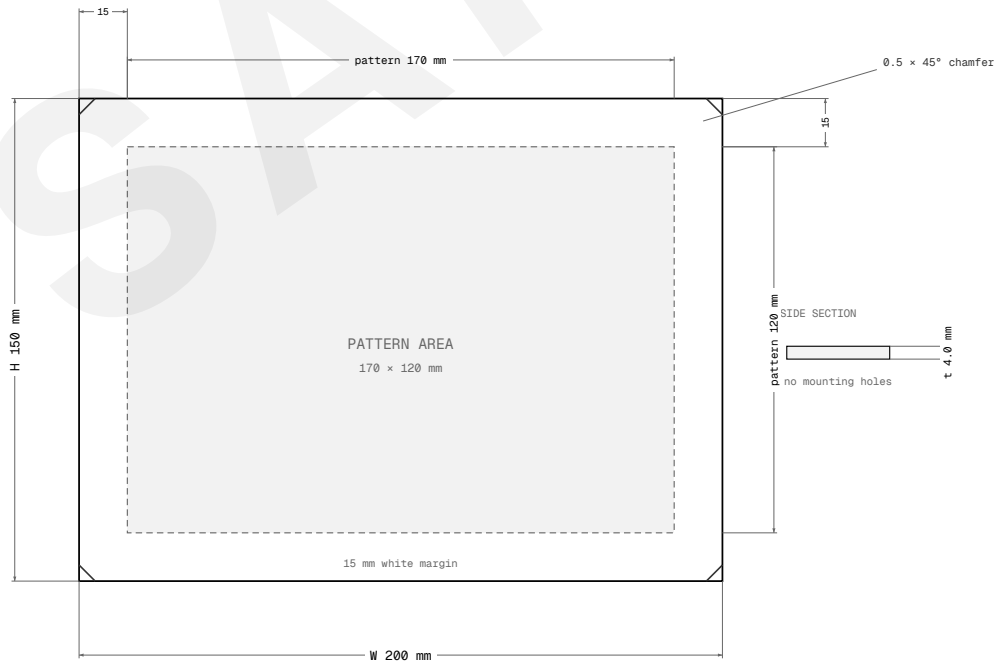
IN THE BOX		CONFIGURATION & SERVICE	
Board	ChArUco ceramic target, selected part number.	Marking on board	Label line with part / pattern / SN / Rev; printed X / Y arrows.
Packaging	Protective carton and handling information.	Lead time	[Lead time — to be decided]
Report	Inspection report and measured CSV for each serialised unit.	Options	Fixture · SVC-X-CERT-ILAC · custom NRE.
Files	`.opencv.json` · `.kalibr.yaml` · `.ros2.yaml` `.halcon.descr` · `.dxf` · measured CSV		

4 Compatibility

SOFTWARE	USE / CAUTION
OpenCV	ChArUco board with `DICT_6X6_250`; this 12-row board follows the OpenCV ≥ 4.6 convention. Do not enable legacy pattern for this board.
Kalibr	No native ChArUco target model. The supplied `.kalibr.yaml` is a geometry reference and needs integration-specific adaptation.
HALCON	Use the supplied descriptor as setup data for the applicable calibration workflow; confirm operator and release compatibility before deployment.
MATLAB	Intended for the Computer Vision Toolbox ChArUco workflow (R2024a or later—confirm release support before use).
ROS 2	Use a `charuco` pattern configuration where supported by the selected `camera_calibration` package; confirm package parameters first.

5 Drawing

200 × 150 mm reference variant



DXF DOWNLOAD

`calib.tools/files/CT-DWG-CHA-C-200150.dxf`

Drawing reference only; use the released DXF for fixture design.

CT-DWG-CHA-C-200150 · Rev A · dimensions nominal unless stated.

SAMPLE

6 Compliance & customs

TOPIC	STATEMENT
RoHS	Materials statement: passive ceramic plate; applicability of Directive 2011/65/EU is to be confirmed.
REACH	[Candidate-list / material statement — to be confirmed for the current supply chain].
HS code	9031 80
Country of origin	[per facts - to be decided]
Packaging	Protective carton, foam interleaf and clean sleeve; packaging configuration to be confirmed.
Handling	FRAGILE · handle by edges; protect the patterned surface from impact and abrasion.

7 Revision history

REV	DATE	CHANGE
A	2026-11-01	Initial issue.

8 Disclaimer

Values are nominal unless stated as measured; per-unit values are in the inspection report CT-IR-<SN>.

SAMPLE